These two additional columns to the inference table are requirements for the Lakehouse Monitor. The `InferenceLog` monitor comes with autogenerated dashboards. However, you need to populate the table. Begin by creating a bronze table for the transaction labels. The Auto Loader is back again, focusing on labels; see `CH7-10-Auto Label Loader` and *Figure 7.25*. In the notebook, you see the creation of the `transaction_labels` table which is very similar to code from *Chapter 3*. In *Figure 7.25*, you use the CDF and CDC Delta features to update the new inference table with the ground truth label.

```
def merge_stream(df, i):
  df.createOrReplaceTempView("labels_cdc_microbatch")

  df._jdf.sparkSession()\
    .sql(f"""MERGE INTO {inference_table} target
        USING
        (SELECT CustomerID, TransactionTimestamp, Label as actual_label
        FROM labels_cdc_microbatch) as source
        ON source.CustomerID = target.CustomerID
        AND source.TransactionTimestamp = target.TransactionTimestamp
        WHEN MATCHED THEN UPDATE SET target.actual_label == source.actual_label
        """)

if spark.catalog.tableExists(inference_table):
  (stream.writeStream
    .foreachBatch(merge_stream)
    .option("checkpointLocation", volume_file_path+inference_table+"/checkpoint_cdc")
    .trigger(availableNow=True)
    .start()
  )
```

Figure 7.25 – The merging of transaction labels into the inference table

You now have a bronze table and the ability to merge new labels into the inference table. However, the inference table is still empty. Let's move forward to create a workflow job, as shown in *Figure 7.26*, to generate predictions every 15 minutes.

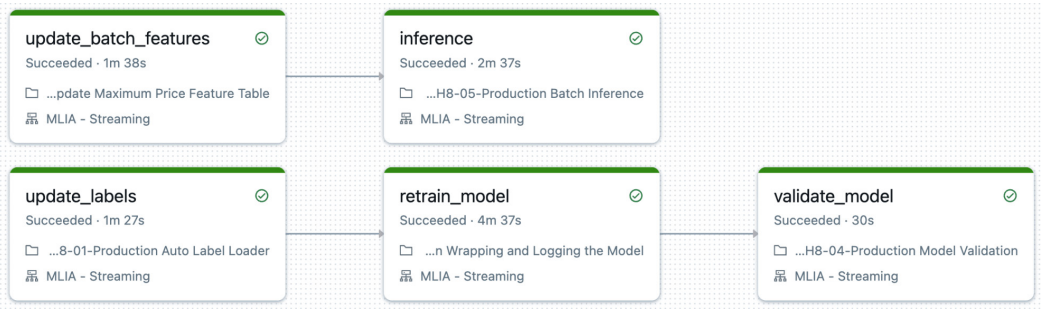


Figure 7.26 – The DAG for the inference (upper) and model retraining (lower) job